Muhammad Ariff Bin Abdul Razak

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PROJECTS

1. Capstone (MET4004) — Acoustic Noise Reduction for Drones — Project Lead — Software & Electronics (Sponsored by KLASS)
Aug 2026 – Apr 2027

- Leading a 3-person team to deliver a proof-of-concept quiet indoor drone for end user MHA to reduce acoustic detectability, targeting a 78 dBA ceiling at a 1 m reference during steady hover, 10-minute hover endurance, and a 370 mm square airframe envelope, with the project proposal approved.
- Owning the software and electronics workstream, selecting a low-noise propulsion stack of 5-inch injection-molded toroidal propellers, custom ducts with acoustic lining and serrated trailing edges, 1300 KV low-RPM motors, and AM32 ESC PWM frequency tuning to shift tonal whine into less perceptible bands.
- Specified the avionics and power architecture (Holybro Pixhawk 6C Mini, PM02 V2 power module, dedicated 150 A power distribution board), yielding a derated 3.3:1 thrust-to-weight ratio at 1,200 g all-up weight to preserve control headroom for manually piloted indoor flight.
- Designed the acoustic validation plan to the Von Karman Institute four-microphone methodology with TU Berlin distance normalization to a 1 m reference, to benchmark directly against HKUST's 90.52 dBA commercial quadcopter baseline and VKI's DJI Phantom 3 SE measurements.
- Specified psychoacoustic post-processing in Python (MoSQITo) under both Zwicker (ISO 532-1) and Moore–Glasberg (ISO 532-2) loudness standards to quantify perceived annoyance beyond raw sound-pressure level.

2. Autonomous Airport Trolley Return Robot (SIT–DigiPen Coursework) — Firmware & Perception

- Built LiDAR-based perception in C++ and ROS for a robot that autonomously returns airport trolleys, developing and integrating obstacle detection, SLAM mapping, and localization with IMU and wheel-odometry sensor fusion on an onboard computer, completing the final demo task.
- Developed RTOS firmware on an STM32F7 microcontroller with concurrent tasks for obstacle detection and joystick control, and a custom serial protocol linking it to ROS for reliable motor control.

WORK EXPERIENCE

Machine Learning & Fullstack Engineer (Intern), KLASS Engineering and Solutions
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, Speech-Forensics) on SLURM-based HPC to assess detection capability against AI-generated media as a public-safety threat, achieving 90.03 AUROC on the NSG-VD replication.
- Ran ablation studies integrating optical flow (RAFT), segmentation (SAM 2), and point tracking (CoTracker) to test whether physical-motion signals in video sequences can distinguish AI-generated content, finding mixed evidence of a usable signal.
- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-sync (LatentSync) models to curate a balanced 1,200-video real/fake dataset, used to compare all four detectors consistently and to test models at small scale before committing larger compute.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits that caught pipeline errors before committing to training and synthetic-deepfake generation runs of up to 5 hours each.
- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for a company-wide internal AI chatbot platform that gave staff a secure alternative to public AI tools, with search across internal documents.

Student Research Engineer, Singapore Institute of Technology (SIT)
Aug 2024 – May 2025

- Designed and built a temperature-controlled enclosure around a compression testing machine for faculty research on how foods behave during chewing, defining the electrical architecture and selecting components before handing the completed chamber over to the research team.
- Engineered closed-loop temperature control with a thermistor and standalone temperature controller, holding the chamber within ±2 °C of a 37 °C body-temperature setpoint for consistent test conditions.

Research Assistant, SUTD – ROAR Lab
Aug 2022 – May 2024 (Volunteer until Nov 2022, Full Time thereafter)

- Led the design of Panthera's intelligent lighting control system to signal the autonomous vehicle's intent to pedestrians and meet testing safety requirements, running a rosserial node on an Arduino under an RTOS that mapped cmd_vel, IMU, and vehicle-state topics to turn, warning, and hazard signals; the platform passed T1 autonomous testing at CETRAN.

- Wired and integrated the 3D LiDAR used for Panthera's obstacle detection, mapping, and localization, verifying sensor output through depth-map visualization, and supported the vehicle through its T1 autonomous field testing.

Trooper & Armourer, Singapore Police Force – Protective Security Command
Jan 2021 – Nov 2022 (Full-Time National Service)

- Performed protective security duties within a deployed team at strategic locations and major national events, operating under strict standard operating procedures.
- Served as unit Armourer alongside trooper duties, managing the issuance, return, and accountability of arms and equipment with zero discrepancies, and maintaining audit-ready records under regulated inventory procedures.

Technology Instructor (Part Time / Full Time), Nullspace Group
Nov 2016 – Present (paused during National Service, Jan 2021 – Nov 2022)

- Instruct students aged 5–17 across multiple schools in STEM-focused courses covering computational thinking, programming (C++, Python), robotics, and microcontrollers as an MOE-certified instructor since age 18, receiving positive feedback from students and schools.

EDUCATION

DigiPen Institute of Technology & SIT
Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic
Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic
Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Internship grade of B+; coursework in Microcontroller Applications, PLCs, and Robotics Integration

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics: Deep Learning, Computer Vision, Multimodal ML, Deepfake Detection, Generative AI, OpenCV, Optical Flow (RAFT), Segmentation (SAM 2), Point-Tracking (CoTracker)
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNNs, DNNs, LSTMs, YOLO, Vision Transformers, Reinforcement Learning, Model Benchmarking, Zero-shot Learning
Robotics, UAV & Autonomous Systems: Embodied AI, ROS, SLAM (3D LiDAR), Perception, Sensor Fusion, Visual-Inertial Odometry (VIO), Autonomous Mobile Robots, Quadcopter Design, UAV Flight Testing, UAV Aeroacoustics, Acoustic Noise Reduction, PX4 Autopilot, ArduPilot, Pixhawk
Embedded, Hardware & Test: Embedded C / C++, RTOS, Microcontrollers (STM32, Arduino), PID Control, Closed-Loop IoT Control, PCB Design, Firmware Development, Hardware Prototyping, Experimental Design, Test & Validation
Software & Fullstack: Python (NumPy, Pandas), C++17, C#, Java, TypeScript, React, Zustand, FastAPI, OpenAPI, WebSockets, Git, Linux/Ubuntu, WSL

AWARDS / ACHIEVEMENTS / CERTIFICATES

- MOE (Ministry of Education) Instructor Certification (Achieved at age 18) — Valid Oct 2018 – Oct 2026 (renewing)

VOLUNTEERING & COMMUNITY SERVICE

ROSCon Singapore 2025 (Open Robotics) — Volunteer
Oct 2025

- Supported event operations, logistics, and attendee experience at ROSCon Singapore 2025, the annual ROS developer conference with ~1,000 participants from 52 countries, including helping run 3 workshops.

SG Makers Against COVID-19

- Initiated a pandemic response project to 3D-print ear savers for SingHealth and the local community, distributing over 10,000 units (Featured on Channel News Asia).